**King Fahd University of Petroleum and Minerals**
**Department of Mathematics**
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# Math 102 Major Exam II Term 223

Original exam date: 31 July 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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**1.** $\int \frac{\sin x}{\cos x+\cos^2 x}\,dx =$

(a) $\ln|6+\tan x|+C$

(b) $\dfrac{6}{\cos^2 x+\cos^3 x}+C$

(c) $\ln|1+\sec x|+C$

(d) $6\ln|\sin x+\cos x|+C$

(e) $\ln|2+\csc x|+C$

**2.** $\int_0^{\pi/2}\cos^3 x\,dx =$

(a) $1$

(b) $\dfrac{3}{4}$

(c) $\dfrac{1}{3}$

(d) $\dfrac{2}{3}$

(e) $\dfrac{1}{2}$

**3.** If $\int \frac{x^2}{x^4-2x^2-8}\,dx = \frac{1}{6}\left[\ln\left|\frac{x-2}{x+2}\right|+\alpha\tan^{-1}\frac{x}{\sqrt{2}}\right]+C$, then $\alpha =$

(a) $\sqrt{3}$

(b) $2\sqrt{3}$

(c) $3\sqrt{2}$

(d) $\sqrt{2}$

(e) $2\sqrt{2}$

**4.** If the region enclosed by $y^2 - 4y = -x$ and $x = 0$ is revolving about the $x$-axis, then the formula for the volume is

(a) $2\pi \int_0^2 4 - (y-2)^2\, dy$

(b) $2\pi \int_0^2 (y^2 - 4y)\, dy$

(c) $\pi \int_0^4 (2 + \sqrt{4-x})^2\, dx$

(d) $2\pi \int_0^4 (2 + \sqrt{4-x})^2\, dx$

(e) $2\pi \int_0^4 y(4y - y^2)\, dy$

**5.** If the volume of the solid generated by revolving the region bounded by the graphs of $y = \dfrac{2}{1+x}$, $y = 0$, $x = 0$, $x = 4$ about the line $y = 4$ is $k\left(\ln 5 - \dfrac{1}{5}\right)$, then $k =$

(a) $8\pi$

(b) $2\pi$

(c) $16\pi$

(d) $4\pi$

(e) $\pi$

**6.** Consider the solid formed by revolving the region bounded by $y = \sqrt{x}$, $y = 0$, $x = 2$, and $x = 4$ about the $x$-axis. The values of $x$, in the interval $[2, 4]$ that divide the solid into three parts of equal volume, are

(a) $2\sqrt{3}$ and $3\sqrt{3}$

(b) $2\sqrt{2}$ and $3\sqrt{5}$

(c) $\sqrt{6}$ and $\sqrt{5}$

(d) $2\sqrt{2}$ and $2\sqrt{3}$

(e) $\sqrt{5}$ and $3$

**7.** $\int x^n \ln x\, dx =$

(a) $\dfrac{x^n}{n+1}\left[(n+1)\ln x - 1\right] + C$

(b) $\dfrac{x^{n+1}}{(n+1)^2}\left[\ln x - 1\right] + C$

(c) $\dfrac{x^n \ln x}{n+1} + C$

(d) $\dfrac{x^{n+1}}{(n+1)^2}\left[x - 1\right] + C$

(e) $\dfrac{x^{n+1}}{(n+1)^2}\left[(n+1)\ln x - 1\right] + C$

**8.** $\int_{\sqrt{3}}^{2} \frac{\sqrt{x^2-3}}{x}\,dx =$

(a) $\pi - \dfrac{\sqrt{3}}{2}$

(b) $1 - \dfrac{\sqrt{3}}{6}\pi$

(c) $\sqrt{3} - \dfrac{\pi}{6}$

(d) $2 - \sqrt{3}\,\pi$

(e) $2\sqrt{3} + \dfrac{\pi}{2}$

**9.** The area of the surface, generated by revolving the curve $y = \sqrt{9 - x^2}$ on $-1 \le x \le 1$ about the $x$-axis, is

(a) $24\pi$

(b) $12\pi$

(c) $112\pi$

(d) $72\pi$

(e) $36\pi$

**10.** $\int \frac{x}{2}\sqrt{x^2+4}\,dx =$

(a) $\dfrac{1}{6}(x^2+4)^{3/2} + C$

(b) $\dfrac{\sqrt{x^2+4}}{2x} + C$

(c) $\dfrac{1}{3}(x^2+4)^{1/2} + C$

(d) $2x\sqrt{x^2+4} + C$

(e) $\dfrac{2}{3}(x^2+4)^{-3/2} + C$

**11.** $\int_{3}^{5} \frac{dx}{\sqrt{x^2-9}} =$

(a) $1$

(b) $\ln 3$

(c) $\ln 2$

(d) $\ln 5$

(e) $\ln 5 - \ln 3$

**12.** $\int_2^4 x \sec^{-1} x \, dx =$

(a) $8 \sec^{-1} 4 + \dfrac{3\sqrt{3} - 3\sqrt{15} - 4\pi}{6}$

(b) $8 \sec^{-1} 4 - \sec^{-1} 2$

(c) $4 \sec^{-1} 4 + 6 \sec^{-1} 2 - \dfrac{\sqrt{3} - \sqrt{5} - 2\pi}{3}$

(d) $\dfrac{\sqrt{3} - \sqrt{5} - \pi}{3} - \sec^{-1} 2$

(e) $\dfrac{\sqrt{3} - \sqrt{5} - 2\pi}{3} + \sec^{-1} 2$

**13.** $\int_1^\infty (1 - x)e^{-x} \, dx$

(a) converges to $-e$

(b) converges to $e$

(c) diverges

(d) converges to $\dfrac{-1}{e}$

(e) converges to $\dfrac{1}{e}$

**14.** $\int \csc^4(3x) \, dx =$

(a) $\dfrac{1}{2} \cot x - \dfrac{1}{3} \cot^3 x + C$

(b) $\dfrac{1}{5} \csc^5(3x) + C$

(c) $-\dfrac{1}{2} \tan(3x) - \dfrac{1}{3} \tan^3(3x) + C$

(d) $-\dfrac{1}{3} \cot(3x) - \dfrac{1}{9} \cot^3(3x) + C$

(e) $-\dfrac{1}{2} \csc(3x) - \dfrac{1}{6} \csc^3(3x) + C$

**15.** The arc length of the graph of $y = \ln(\sin x)$ over $\left[\dfrac{\pi}{4}, \dfrac{3\pi}{4}\right]$ equals to

(a) $\int_{\pi/4}^{3\pi/4} \cot x \, dx$

(b) $\int_{\pi/4}^{3\pi/4} \csc x \, dx$

(c) $\int_{\pi/4}^{3\pi/4} \sec x \, dx$

(d) $\int_{\pi/4}^{3\pi/4} \tan x \, dx$

(e) $\int_{\pi/4}^{3\pi/4} \ln(\cos^2 x) \, dx$

**16.** The volume of the solid, generated by revolving the region bounded by the graphs of the equations $y = \tan x$, $y = 0$, $x = \dfrac{-\pi}{4}$, $x = \dfrac{\pi}{4}$ about the $x$-axis, is

(a) $2\pi\left(2 - \dfrac{\pi}{4}\right)$

(b) $2\pi(1 - \pi)$

(c) $\pi\left(1 - \dfrac{\pi}{4}\right)$

(d) $2\pi\left(1 - \dfrac{\pi}{2}\right)$

(e) $\pi\left(2 - \dfrac{\pi}{2}\right)$

**17.** $\int (\ln x)^2\, dx =$

(a) $(\ln x)^2 - 2\ln x - x + C$

(b) $x(\ln x)^2 - x^2 + C$

(c) $x\left[(\ln x)^2 - 2\ln x + 2\right] + C$

(d) $\dfrac{(\ln x)^2 - \ln x - x}{2} + C$

(e) $\dfrac{(\ln x)^2}{2} + C$

**18.** The volume of the solid, generated by revolving the region bounded by the graph of $y = x^3$, $y = 0$, $x = 2$ about the line $x = 4$, is

(a) $\dfrac{\pi}{5}$

(b) $\dfrac{88\pi}{5}$

(c) $\dfrac{96\pi}{5}$

(d) $\dfrac{116\pi}{5}$

(e) $\dfrac{32\pi}{5}$

# Answer Key

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| Q | Ans | Brief explanation |
|---|---|---|
| 1 | **(c)** | $u = \cos x$: $-\int \frac{du}{u(1+u)} = \ln\left\lvert\frac{1+u}{u}\right\rvert = \ln\lvert 1+\sec x\rvert + C$. |
| 2 | **(d)** | $\int_0^1 (1-u^2)\,du = \frac{2}{3}$ with $u = \sin x$. |
| 3 | **(d)** | $\frac{2}{3}\frac{1}{x^2-4} + \frac{1}{3}\frac{1}{x^2+2}$ gives $\frac{1}{6}[\ln\lvert\frac{x-2}{x+2}\rvert + \sqrt{2}\tan^{-1}\frac{x}{\sqrt{2}}]$. |
| 4 | **(e)** | $x = 4y - y^2$, $y \in [0,4]$: shells give $V = 2\pi\int_0^4 y(4y-y^2)\,dy$. |
| 5 | **(c)** | Washers: $V = \pi\int_0^4 [16 - (4 - \frac{2}{1+x})^2]dx = 16\pi(\ln 5 - \frac{1}{5})$, so $k = 16\pi$. |
| 6 | **(d)** | Total $6\pi$; $\pi\frac{a^2-4}{2} = 2\pi, 4\pi$ give $x = 2\sqrt{2}$ and $2\sqrt{3}$. |
| 7 | **(e)** | By parts, $u = \ln x$, $dv = x^n dx$: $\frac{x^{n+1}}{(n+1)^2}[(n+1)\ln x - 1] + C$. |
| 8 | **(b)** | $x = \sqrt{3}\sec\theta$: $\sqrt{3}\int_0^{\pi/6}\tan^2\theta\,d\theta = 1 - \frac{\sqrt{3}}{6}\pi$. |
| 9 | **(b)** | $y\sqrt{1+y'^2} = 3$, so $S = 2\pi\int_{-1}^1 3\,dx = 12\pi$. |
| 10 | **(a)** | $u = x^2+4$: $\frac{1}{4}\int u^{1/2}du = \frac{1}{6}(x^2+4)^{3/2} + C$. |
| 11 | **(b)** | $\ln(x + \sqrt{x^2-9})\Big\rvert_3^5 = \ln 9 - \ln 3 = \ln 3$. |
| 12 | **(a)** | By parts: $\left[\frac{x^2}{2}\sec^{-1}x - \frac{1}{2}\sqrt{x^2-1}\right]_2^4 = 8\sec^{-1}4 + \frac{3\sqrt{3}-3\sqrt{15}-4\pi}{6}$. |
| 13 | **(d)** | $\int(1-x)e^{-x}dx = xe^{-x}$; the limit gives $0 - e^{-1} = -\frac{1}{e}$. |
| 14 | **(d)** | $u = \cot 3x$: $-\frac{1}{3}\int(1+u^2)du = -\frac{1}{3}\cot 3x - \frac{1}{9}\cot^3 3x + C$. |
| 15 | **(b)** | $y' = \cot x$, $\sqrt{1+\cot^2 x} = \csc x$: $L = \int_{\pi/4}^{3\pi/4}\csc x\,dx$. |
| 16 | **(e)** | Disks: $V = 2\pi\int_0^{\pi/4}\tan^2 x\,dx = 2\pi(1-\frac{\pi}{4}) = \pi(2-\frac{\pi}{2})$. |
| 17 | **(c)** | Parts twice: $x(\ln x)^2 - 2x\ln x + 2x + C$. |
| 18 | **(c)** | Shells: $V = 2\pi\int_0^2 (4-x)x^3\,dx = \frac{96\pi}{5}$. |